

Understanding Regression

Regression is a statistical method used to model the relationship between a **dependent variable** (the outcome you want to predict) and one or more **independent variables** (the features or predictors).

While classification predicts "categories" (like Spam vs. Not Spam), regression predicts **continuous numerical values**, such as house prices, temperature, or stock market trends.

Simple Linear Regression (SLR)

Simple Linear Regression is the most basic form of regression. It models the relationship between a single independent variable x and a single dependent variable y by fitting a straight line through the data points.

The Linear Equation

The mathematical formula for the regression line is:

$$y = \beta_0 + \beta_1 x + \epsilon$$

- y : The predicted value (Dependent Variable).
- x : The input value (Independent Variable).
- β_0 (**Intercept**): The value of y when $x = 0$.
- β_1 (**Slope**): The change in y for every one-unit change in x .
- ϵ (**Error Term**): The difference between the actual data points and the predicted line (noise).

How it is Calculated: Ordinary Least Squares (OLS)

To find the "best" line, we need to determine the values of β_0 and β_1 that minimize the distance between the actual data points and the line. The most common method is **Ordinary Least Squares (OLS)**.

OLS minimizes the **Sum of Squared Errors (SSE)**. The formulas for the parameters are:

1. **Slope (β_1):**

$$\beta_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

2. **Intercept (β_0):**

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

(Where $\bar{x}$ and $\bar{y}$ are the means of x and y respectively.)

Errors and Residuals

An **error** (or residual) is the vertical distance between an actual observed value (y) and the value predicted by the model ($\hat{y}$).

$$e_i = y_i - \hat{y}_i$$

- **Positive Residual:** The actual value is above the prediction.
- **Negative Residual:** The actual value is below the prediction.
- **Goal:** We square these errors so that positive and negative errors don't cancel each other out during calculation.

Evaluation Metrics

How do we know if our model is any good? We use these standard metrics:

Metric
Mean Absolute Error (MAE)
Mean Squared Error (MSE)
Root Mean Squared Error (RMSE)
R-Squared (R^2)

Step-by-Step Example (Manual Calculation)

Suppose we want to predict **Salary** based on **Years of Experience**.

Experience (x)	Salary (y)
1	45,000
2	50,000
3	60,000

Step 1: Calculate Means

- $\bar{x} = (1 + 2 + 3)/3 = 2$
- $\bar{y} = (45 + 50 + 60)/3 = 51.66$

Step 2: Calculate β_1 (Slope)

- Numerator: $(1 - 2)(45 - 51.66) + (2 - 2)(50 - 51.66) + (3 - 2)(60 - 51.66) = 6.66 + 0 + 8.34 = 15$
- Denominator: $(1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2 = 1 + 0 + 1 = 2$
- $\beta_1 = 15/2 = 7.5$ (Salary increases by 7,500 per year)

Step 3: Calculate β_0 (Intercept)

- $\beta_0 = 51.66 - (7.5 \times 2) = 36.66$
- **Final Model:** $y = 36.66 + 7.5x$

Implementation in Python

We typically use `scikit-learn` for this, as it handles the heavy lifting.

Python

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# 1. Prepare Data
X = np.array([[1], [2], [3], [4], [5]]) # Experience
y = np.array([45000, 50000, 60000, 65000, 75000]) # Salary

# 2. Initialize and Train Model
```

```
model = LinearRegression()
model.fit(X, y)

# 3. Make Predictions
y_pred = model.predict(X)

# 4. Evaluation
print(f"Slope (Beta1): {model.coef_[0]}")
print(f"Intercept (Beta0): {model.intercept_}")
print(f"R-squared: {r2_score(y, y_pred):.2f}")

# 5. Visualization
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X, y_pred, color='red', label='Regression Line')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.legend()
plt.show()
```